Lecture 4

Errors in digital trace data collection

Laura Boeschoten
Associate Professor
Dept. of Methodology, Statistics and Data Science

Goals of this lecture

1. **Understand** the concept of an error and the purpose of error frameworks
2. **Distinguish** between different types of errors
3. **Identify** different errors in study designs
4. **Apply** enriched and amplified asking approaches and understand the role of record linkage here

Literature

- **Chapter 3:** “Asking questions” in *Bit by bit: Social science in the digital age (2017),* Matthew Salganik 3.1 – 3.4 & 3.6

- **Chapter 4:** “Errors of nonobservation: Sampling and coverage” in *Data collection with Wearables, Apps and Sensors (2023)*, Florian Keusch, Bella Struminskaya, Stephanie Eckman & Heidi Guyer

- **Corten et al., (2024)** Assessing Mobile Instant Messenger Networks with Donated Data.

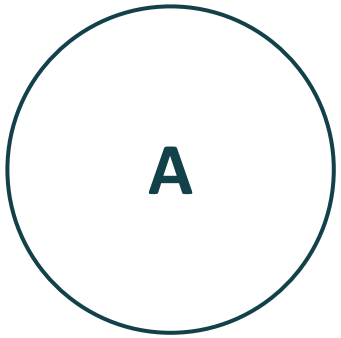
- **Chapter 3:** “Record Linkage” in Big data and Social Science 3.1 & 3.2

Outline

1. Relevance of survey methodology
2. Errors in digital trace data
3. Combining survey and digital trace data

01

Relevance of survey methodology



Remember DTD definitions from week 1

Readymade, custom made

DEFINITION

Readymade

Repurpose big data sources that were originally created by companies and governments.

DEFINITION

Custom made

A researcher started with a specific question and then used the tools of the digital age to create the data needed to answer that question.

See Salganik (2018), Bit by Bit: Social Research in the Digital Age

Found data, designed data

From a researcher's perspective, big data sources look **“found”**: already out there, waiting to be used. But someone **“designed”** it first.

DEFINITION

Found data

Data that already exists “out there”, which researchers didn't set out to create. Examples are platform logs, transaction records, social media posts. It was already there, before it was collected for a (research) purpose.

DEFINITION

Designed data

The same data, seen from another angle: every field, format, and default was chosen by someone (e.g. a platform, a research team) for a reason of their own. This means that no aspect of the data is neutral or automatic.

See Salganik (2018), Bit by Bit: Social Research in the Digital Age

B

Introduction to survey methodology

What is survey methodology?

Description

- It’s a research field in applied statistics.
- It is concerned with how surveys are designed and conducted to study populations.
  - How are individual units sampled from a population?
  - How can procedures be improved to increase both the quality and the quantity of responses?
- This expertise is important because the reliability of any statistical inference about a population depends directly on how the underlying questions and sample were constructed.

Some survey history

Table 3.1: Three Eras of Survey Research Based on Groves (2011)

	Sampling	Interviewing	Data environment
First era	Area probability sampling	Face-to-face	Stand-alone surveys
Second era	Random-digit dialing (RDD) probability sampling	Telephone	Stand-alone surveys
Third era	Non-probability sampling	Computer-administered	Surveys linked to big data sources

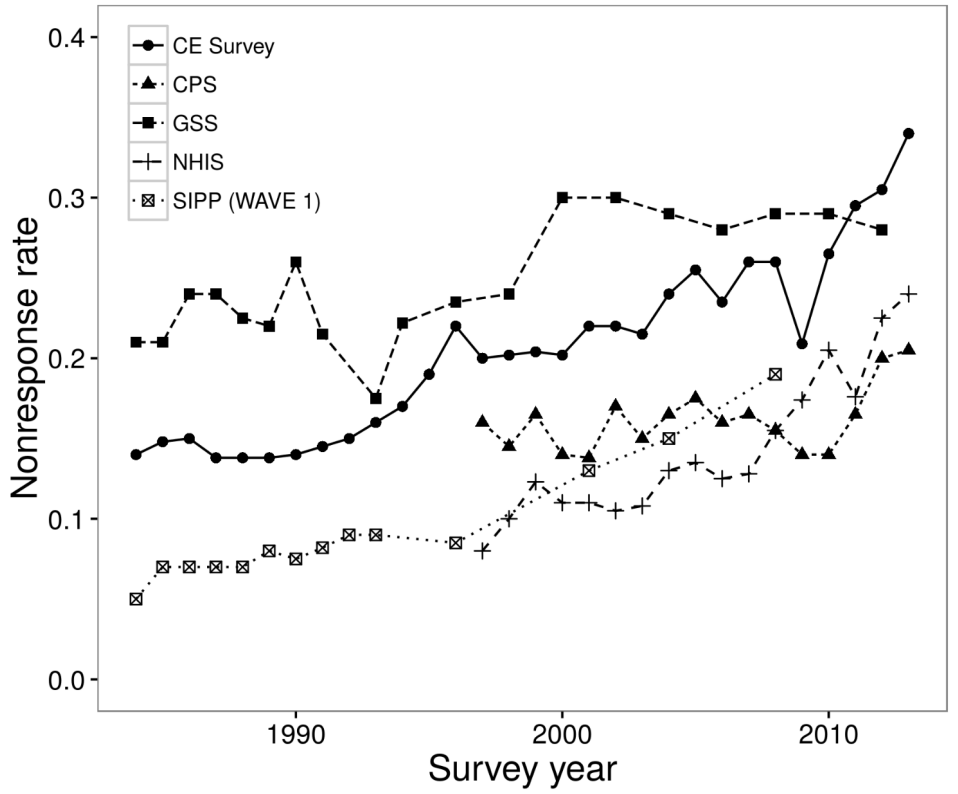
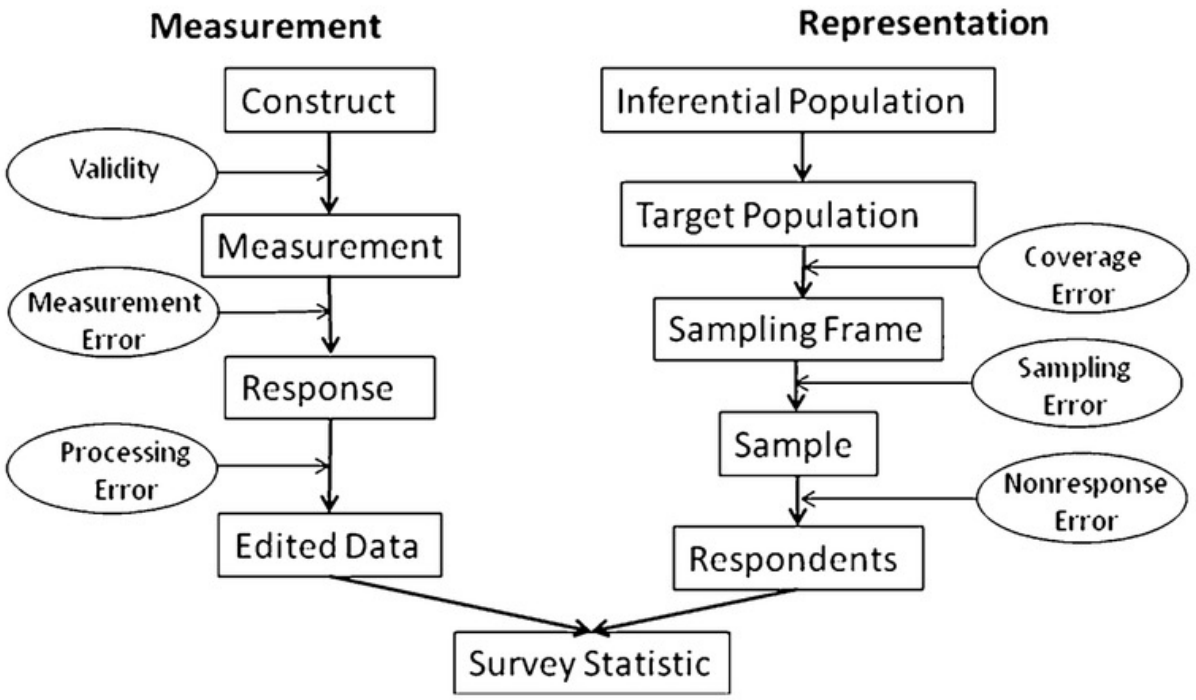


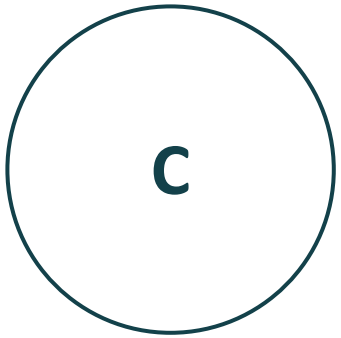
Figure 3.5: Nonresponse has been increasingly steadily, even in high-quality expensive surveys (National Research Council 2013; B. D. Meyer, Mok, and Sullivan 2015). Nonresponse rates are much higher for commercial telephones surveys, sometimes even as high as 90% (Kohut et al. 2012). These long-term trends in nonresponse mean that data collection is more expensive and estimates are less reliable. Adapted from B. D. Meyer, Mok, and Sullivan (2015), figure 1.

Total Survey Error Framework

In each step of the design and analysis phase of a survey, errors can arise that affect the quality of the final statistic of interest.



Source: Groves et al. (2009)



Measurement error

Why study measurement error?

- Measurement error can bias means, but it especially biases relationships.
- This includes studying change over time.
- That is why it is important to know how much measurement error there is.

Measurement error – definition

DEFINITION

Measurement error: the difference between a respondent's answer and the true value.

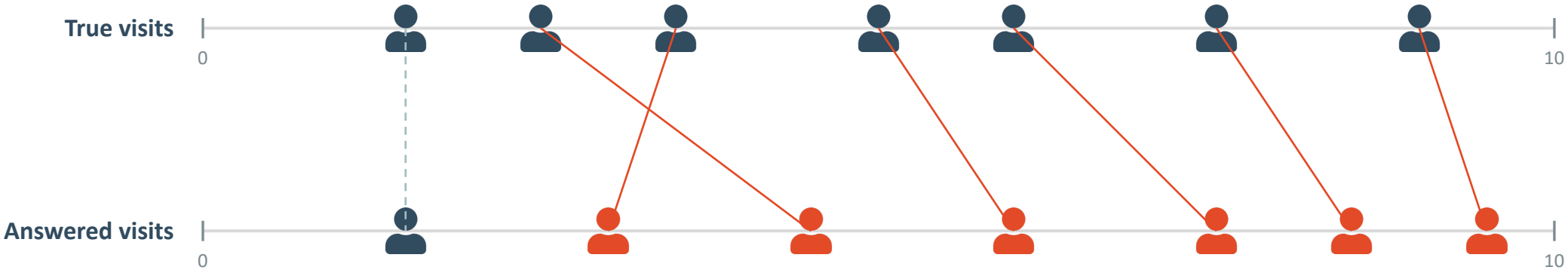
EXAMPLE

Mr. Jones says he visited the doctor three times, but he actually went four times (he forgot a follow-up visit). The measurement error is –1.

Observed value = True value + Error

The answer you observe is shaped by things other than the value you're after. Those things are defined as "errors."

How often have you visited the doctor in the past month?



Most respondents report a different number of visits than they actually made, some are off by a little, some by a lot. This creates measurement error.

EXERCISE

By eyeballing the diagram on the previous slide, what's your guess for the mean of the true visits? And of the answered visits?

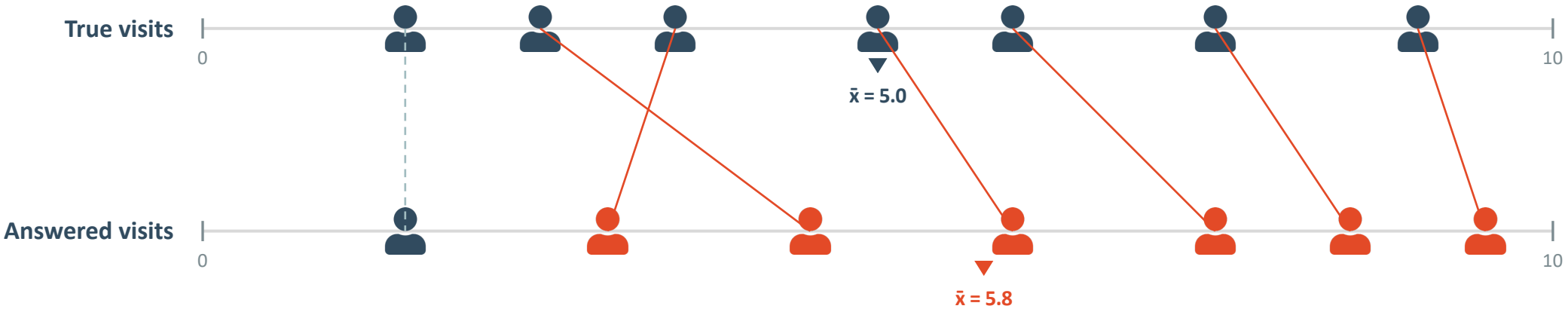
What about the variance: which row looks more spread out?

How often have you visited the doctor in the past month?

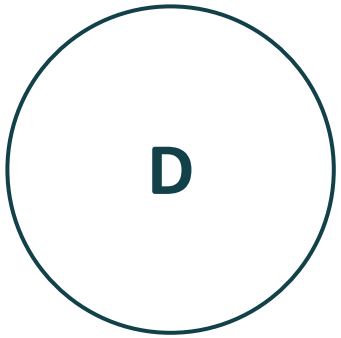
ANSWER: the observed distribution (bottom) differs from the true distribution (top), resulting in bias in the:

Mean: 5.0 → 5.8

Variance: 6.3 → 7.4



Adapted from Salganik (2017), Bit by Bit — Ch. 3



Representation error

How often have you visited the doctor in the past month?

- Imagine the answer to this question is related to something else. For example, older people visit the doctor more often.
- If we ask this question in an internet survey, some older respondents have no computer and as a result are never observed at all.



dashed = did not respond (no internet access)

EXERCISE

By eyeballing the diagram on the previous slide, what's your guess for the mean visits across everyone? And across just the people who responded?

What about the variance, does it get bigger or smaller once the non-responders drop out?

How often have you visited the doctor in the past month?

ANSWER: older non-respondents are systematically missing from the achieved sample, resulting in bias in the:

Mean: 5.0 → 3.7

Variance: 6.3 → 2.7

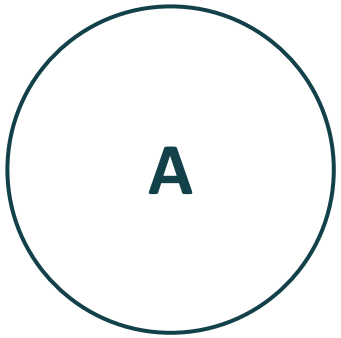


dashed = did not respond (no internet access)

Adapted from Salganik (2017), Bit by Bit — Ch. 3

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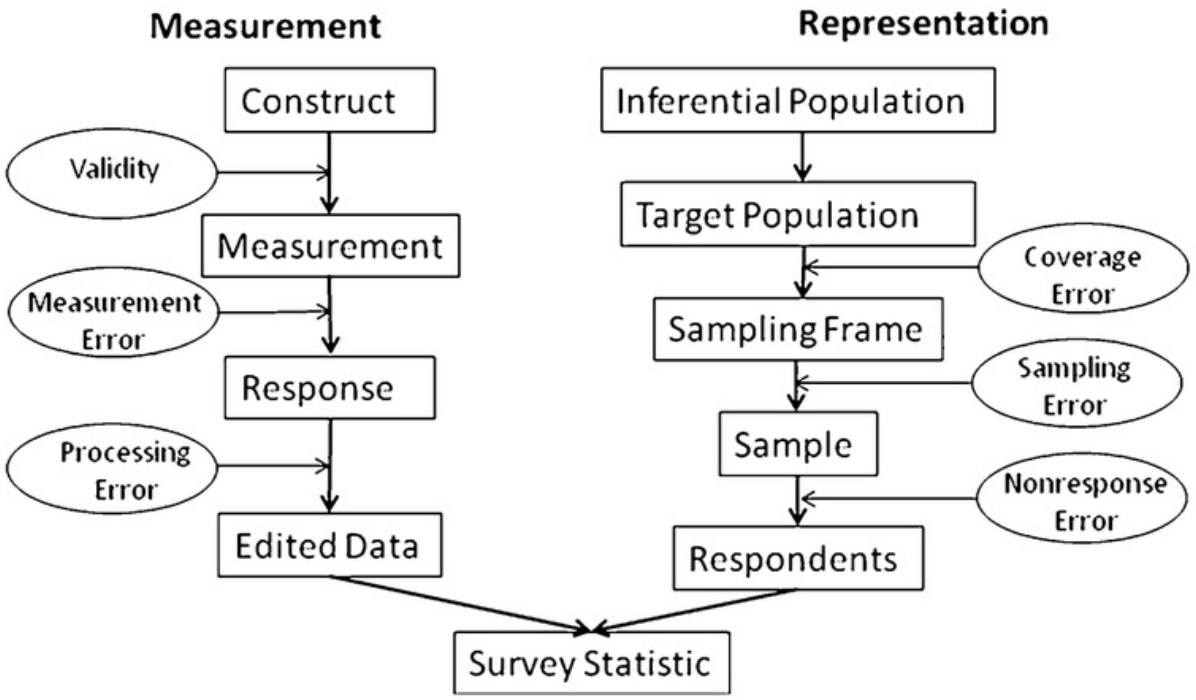
Errors in digital trace data



Error frameworks

Total Survey Error Framework

In each step of the design and analysis phase of a survey, errors can arise that affect the quality of the final statistic of interest.

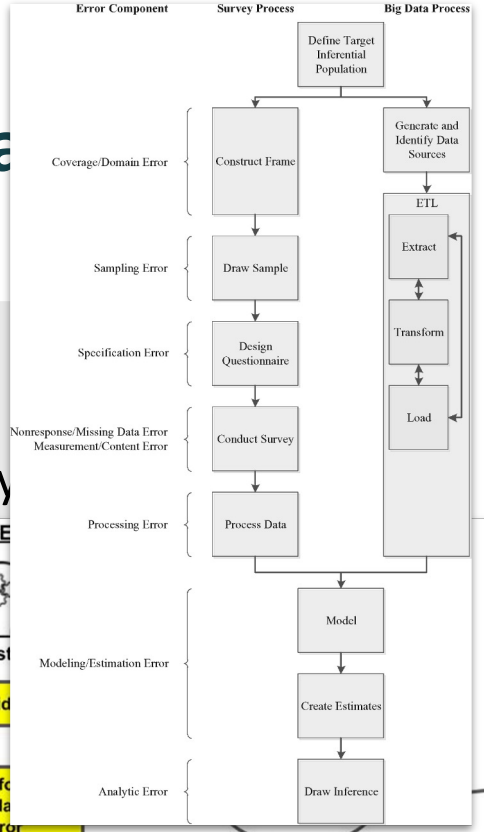


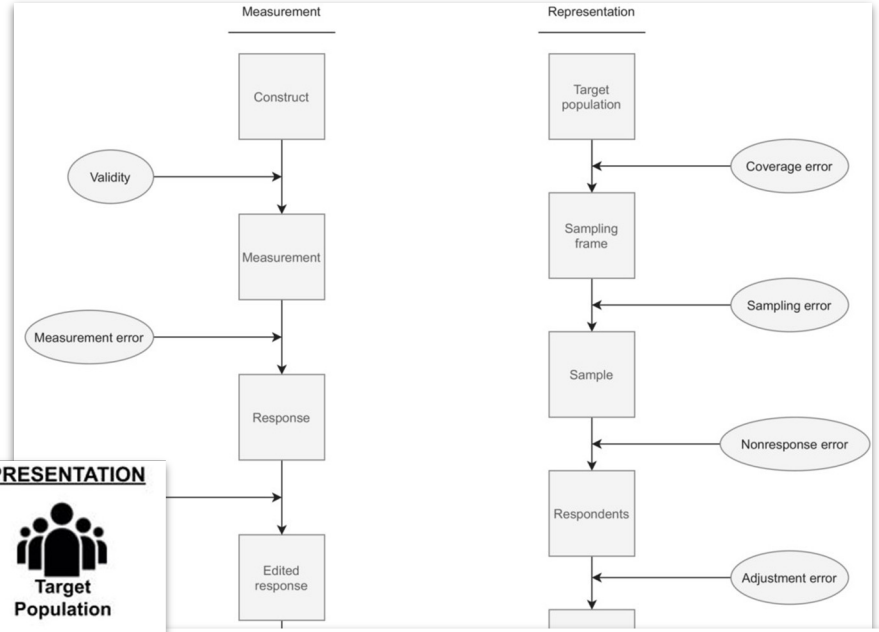
Source: Groves et al. (2009)

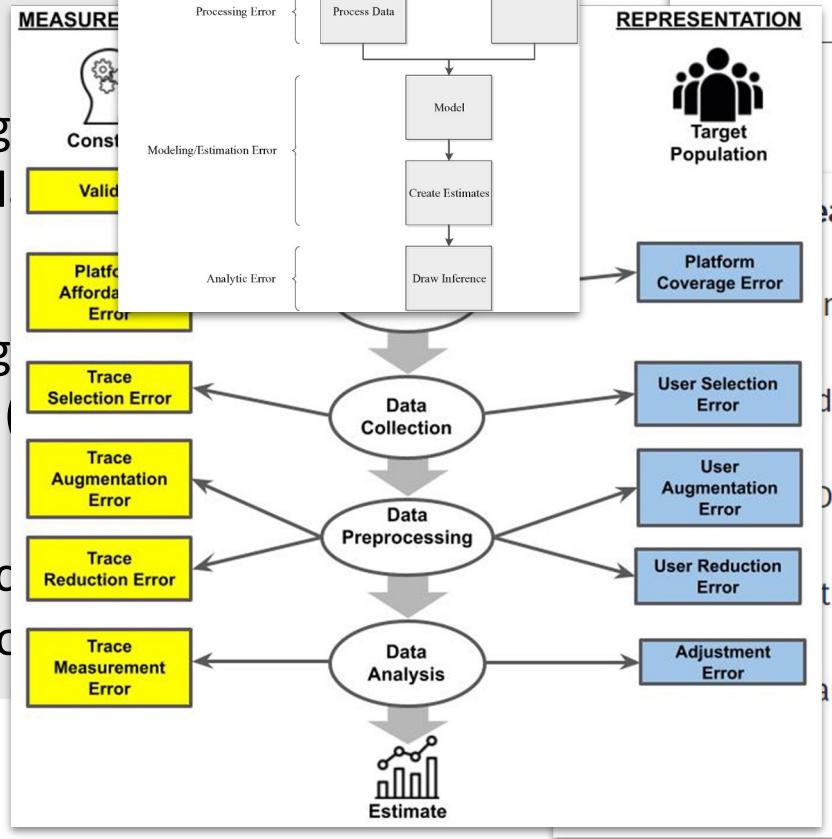
Error frameworks for digita

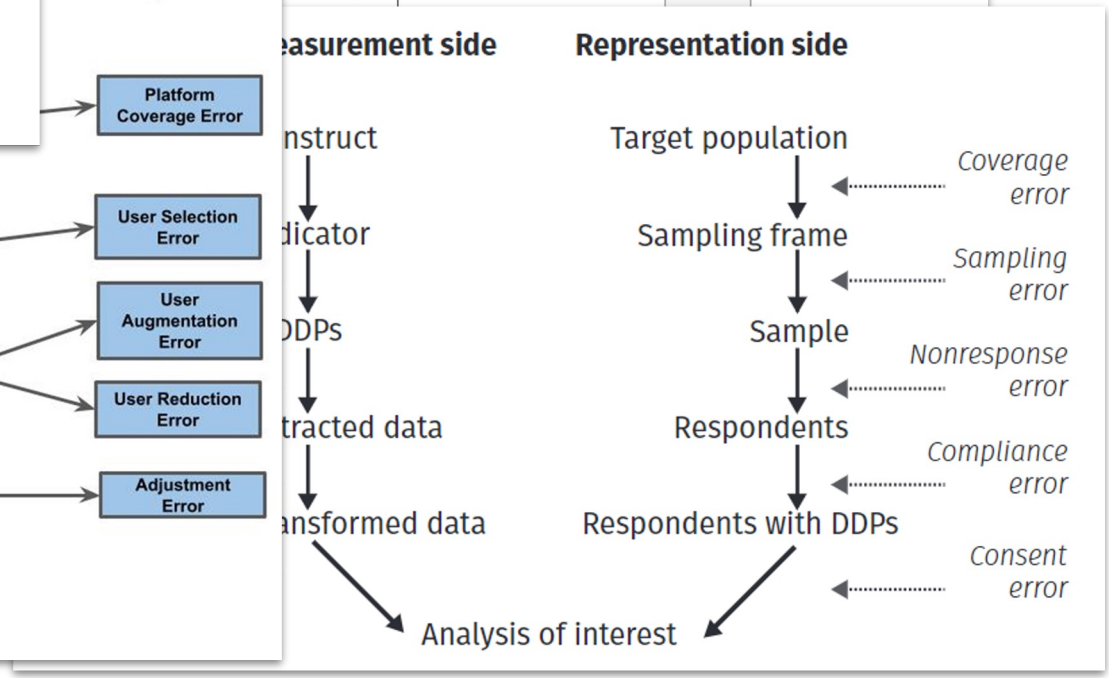
Examples

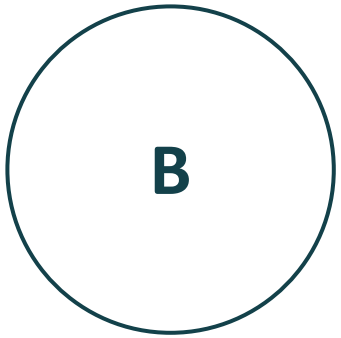
- Total Error Framework (TEF) (Amay 2020)
- Total Error Framework for Dig Human Behavior on Online Pl On) (Sen et al. 2021)
- Total Error Framework for Dig Collected with Meters (TEM) (2022)
- Total Error for Social Scientific with DDPs (Boeschoten, Auslo











Problems with representation

Problems with representation in surveys



Adapted from Salganik (2017), Bit by Bit — Ch. 3, Fig 3.2

Problems with representation in digital trace data

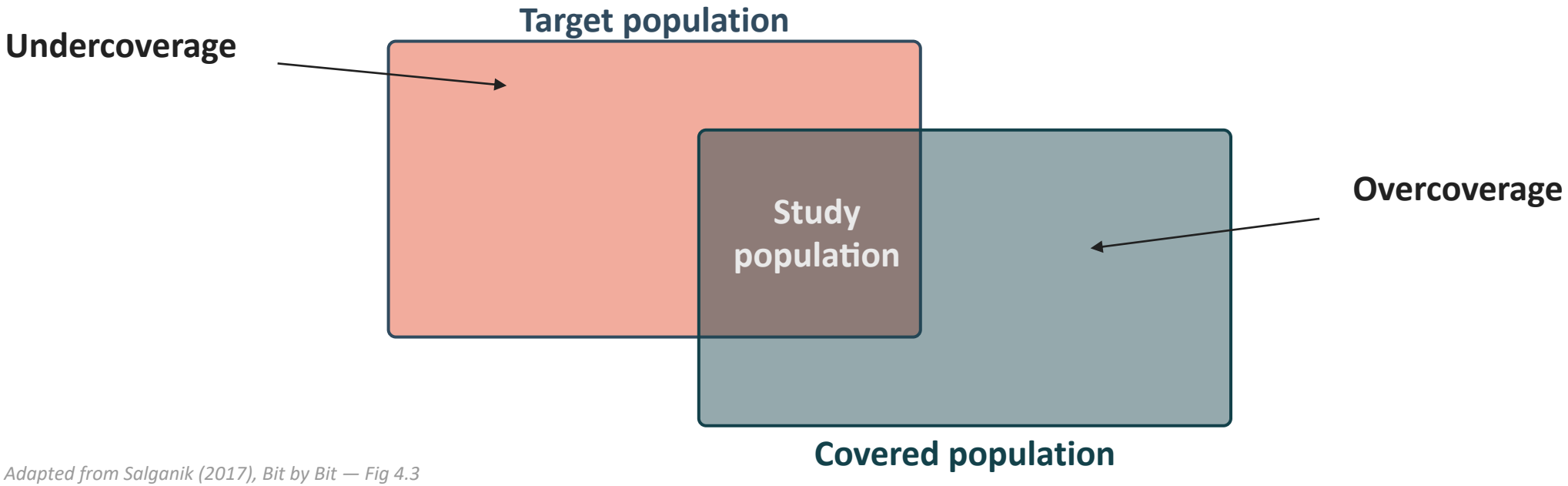
1. **Coverage:** Who uses the platform you're studying in the first place?

2. **Sampling:** Who actually receives your invite to participate?

3. **Non-response:** Who is willing to share their data, and who declines?

Coverage error

Coverage error: the difference between the target population and the population you can actually reach through a platform or sensor



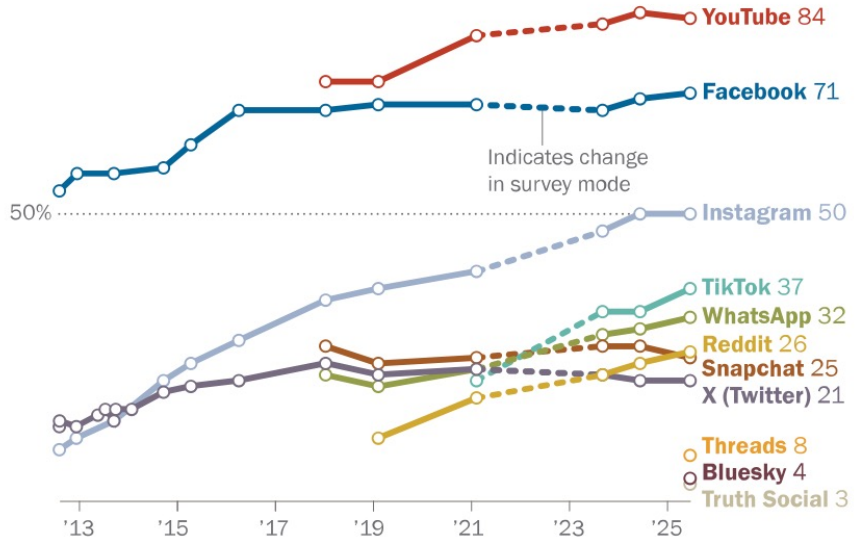
Adapted from Salganik (2017), Bit by Bit — Fig 4.3

Coverage error in practice

- Not everyone uses platforms: non-users have a 0% chance of ever being included.
- Overcoverage happens too: someone running multiple accounts on the same platform can look like several independent people in your data.

TikTok, Instagram, WhatsApp and Reddit have continued to gain users in recent years

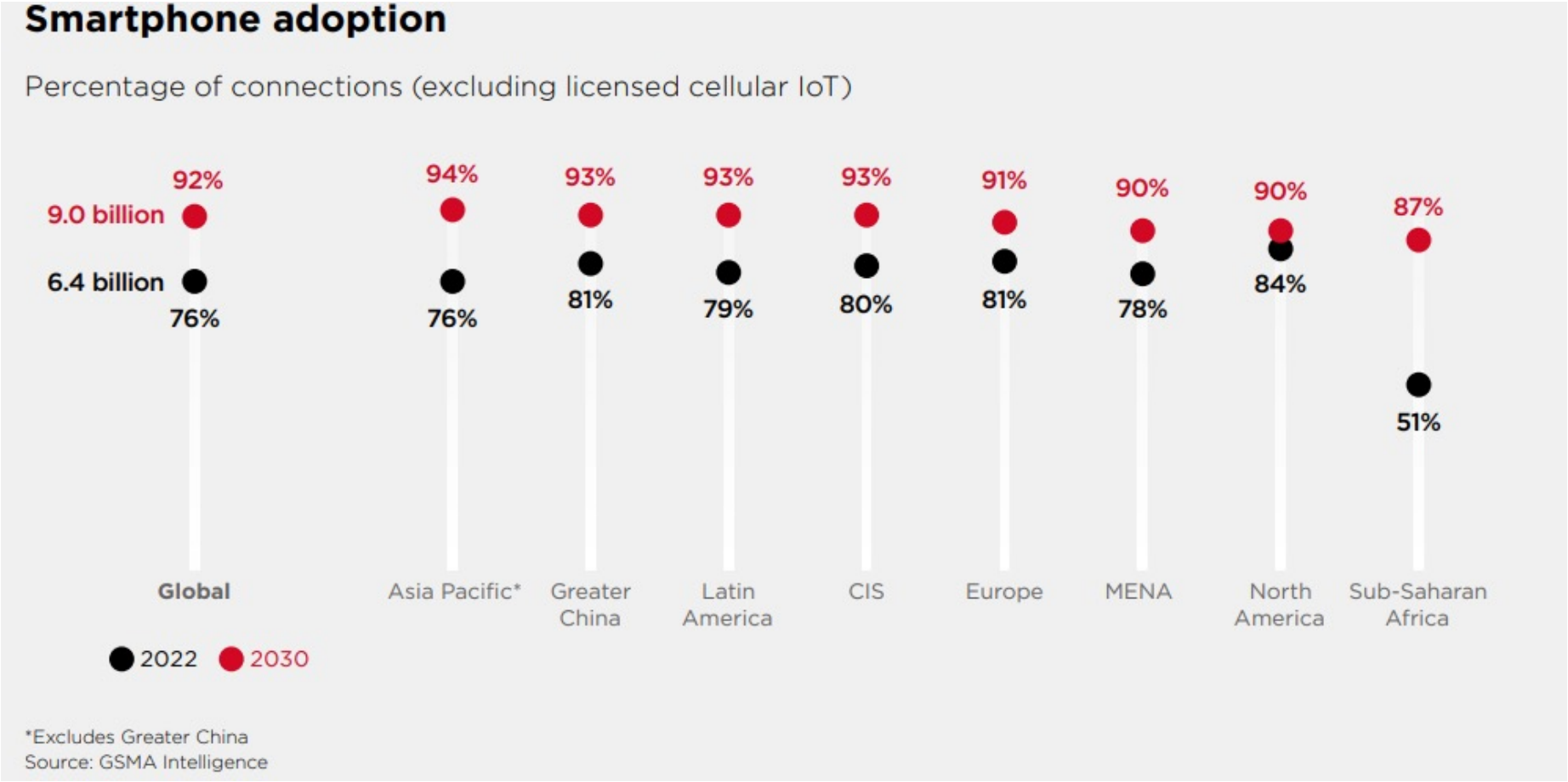
*% of U.S. adults who say they **ever** use the following*



Note: The dotted line indicates a change in mode. Polls from 2012 to 2021 were conducted via phone. In 2023, the poll was conducted via web and mail. In 2024 and 2025, the poll was conducted via web, mail and phone. Refer to the topline for more information on how question wording varied over the years. Not all platforms were asked about in the full 2013-2025 timespan. Those who did not give an answer are not shown.
Source: Survey of U.S. adults conducted Feb. 5-June 18, 2025.
"Americans' Social Media Use 2025"

PEW RESEARCH CENTER

Coverage error in practice

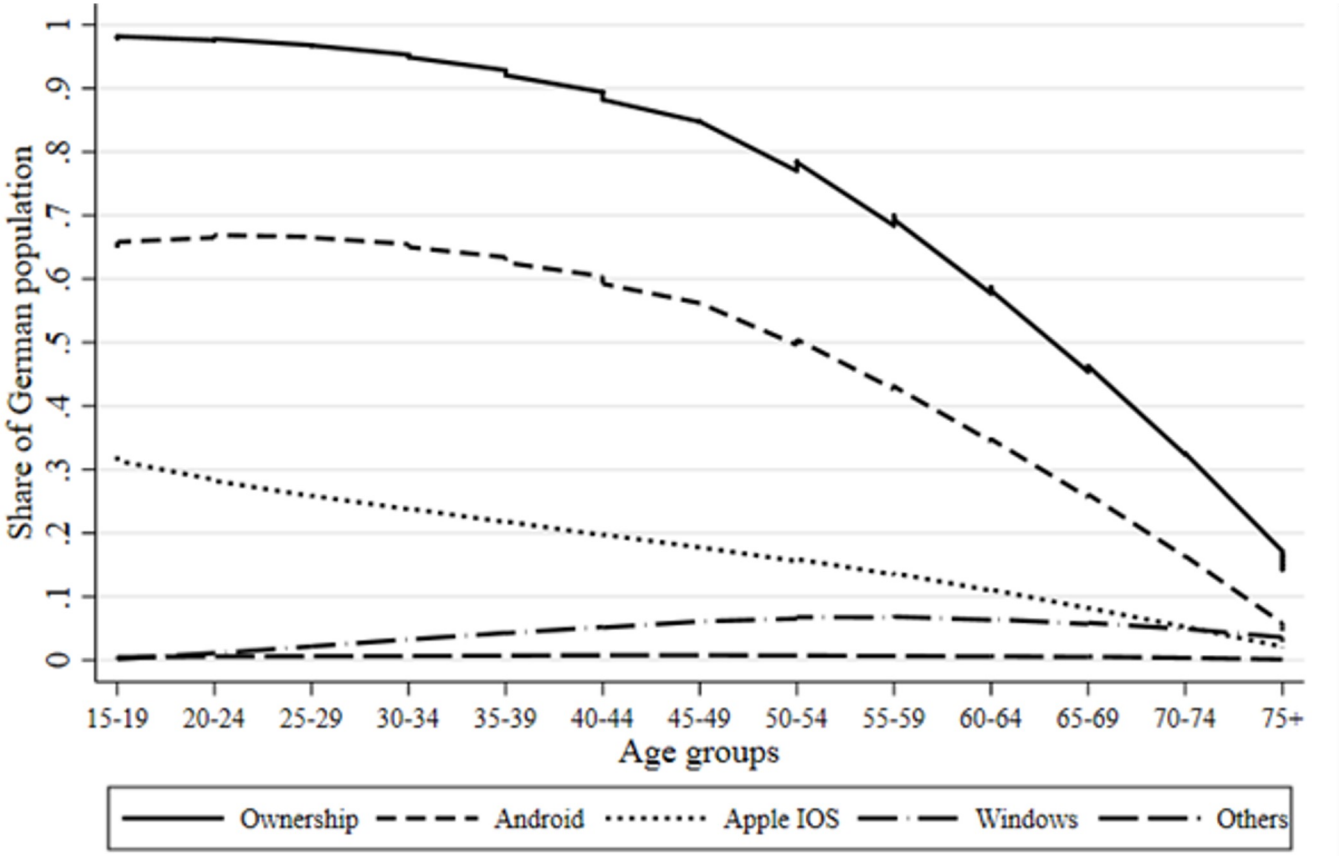


Coverage error in practice

- Smartphone ownership correlates with age, education, nationality, region, and community size (Keusch et al. 2020, Germany).

Bias from ownership gaps is small for most substantive measures, **but substantial for iPhone ownership specifically.**

Source: Keusch et al. (2020); PASS Wave 11, n = 13,703, Germany



Sampling error

Sampling error: a statistical error that occurs when the analyst selects a sample that isn't representative of the population being studied.

- We only observe a subset of the population, not the entire group. Even a perfectly random sample will not match the population exactly.
- Larger samples have smaller sampling error by design.
- DTD examples: Your web advertisement gets blocked by certain browsers.

investopedia.com/terms/s/samplingerror.asp

1 Coverage

2 Sampling

3 **Non-response**

Non-response error

Who is willing to share their data, and who declines? The literature uses several related terms:

Non-participation

Non-willingness

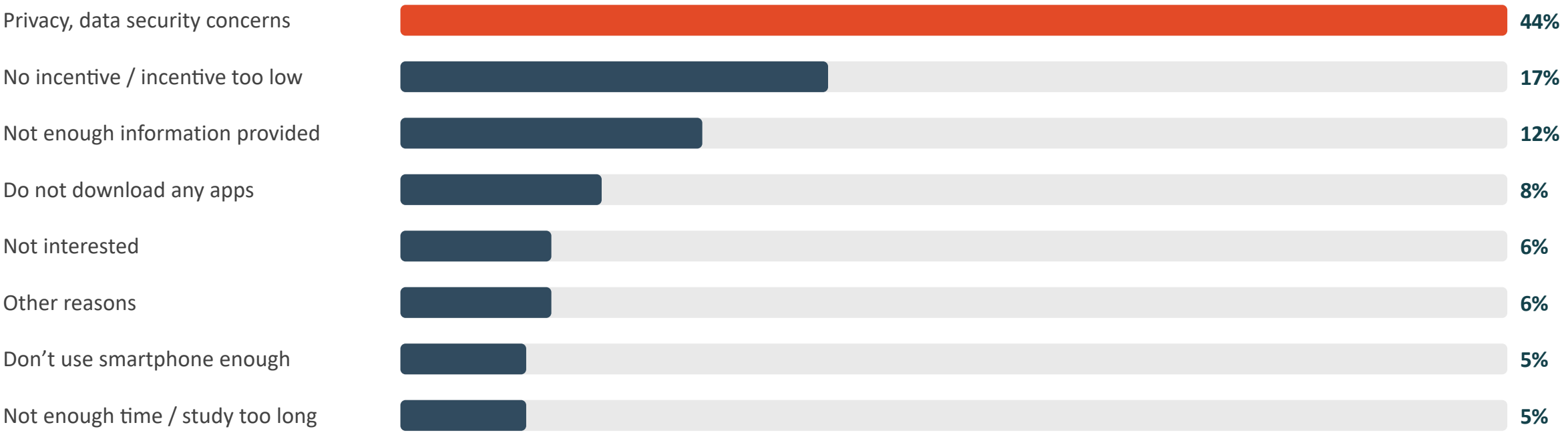
Non-compliance

Nonresponse error: someone who was selected to participate in a study doesn't provide the requested data, because they decline or don't complete it.

Keusch et al. (2019); Keusch, Bähr, et al. (2022)

Why people say no

Reasons not to participate in research app studies:



n = 1,154; Keusch et al. (2019)

Other mechanisms of non-response

Agency
Willingness is higher when participants control the data collection themselves.

Framing
Emphasizing benefits can increase willingness to participate.

Prior experience
Having downloaded a research app before increases willingness.

Sponsor
A university sponsor gets higher willingness than market research or a statistical office.

Smartphone skills
More smartphone activities (GPS, photos, banking) → higher willingness.

Sociodemographics
Education and age both correlate with willingness to participate.

Keusch et al. (2019, 2021, 2022); Struminskaya et al. (2020, 2021); Revilla et al. (2019); Wenz et al. (2019); Wenz & Keusch (in press); Jäckle et al. (2019); McCool et al. (2021)

Willingness and compliance in practice

Google Semantic Location History, donated using local processing, Dutch online panel (CentERpanel), N = 1,035, 75% AAPOR RR1

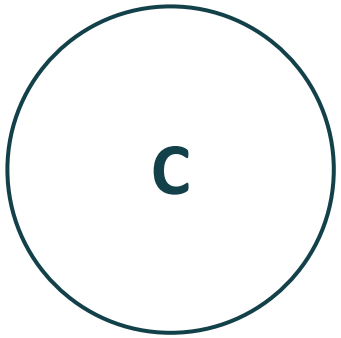
30%
willing to donate

14%
eventually donated

5.5%
understood every part of the consent request

People who understood the consent request better were more likely to be willing and to actually donate *(OR = 1.57–1.80, both p < .001).*

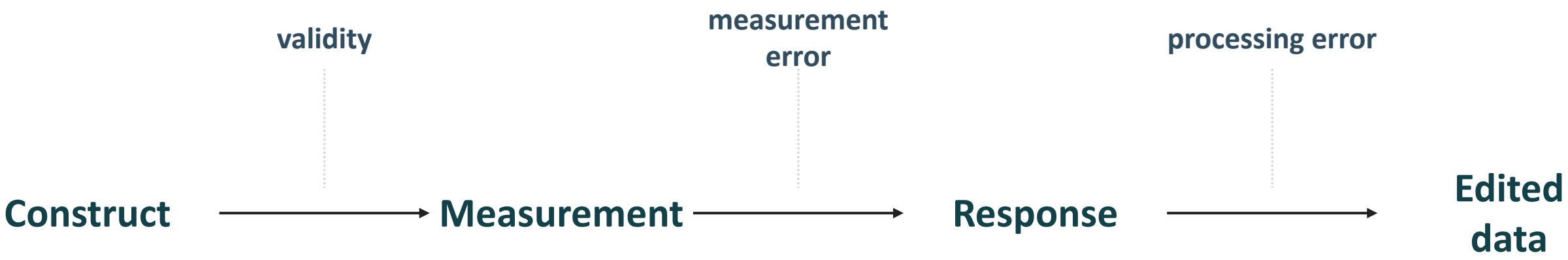
Struminskaya & Boeschoten (2022)



Problems with measurement

Problems with measurement in surveys

The traditional survey approach



Groves et al.; Total Survey Error framework, measurement side

Problems with measurement in digital trace data

1. **Validity:** Are you measuring what you actually want to measure?

2. **Measurement error:** Are your measurements correct? On the platform, or in your own app/plug-in/tool?

3. **Processing error:** What do you need to do to turn the raw data into the measurement you're interested in?

Validity

Validity: are you measuring the concept you are really interested in, or just something that is correlated with it?

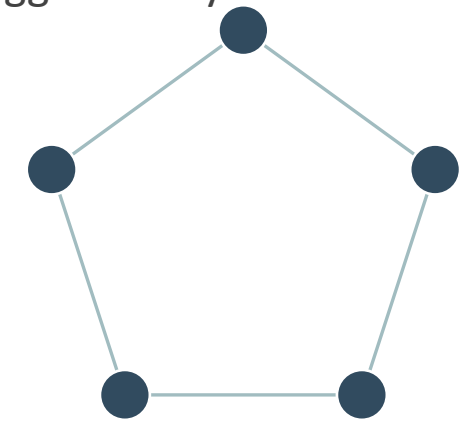
EXAMPLE

- We're interested in someone's mood.
- We use "facial expression in photos" as an indicator, collected through data donation.
- **Are we measuring the concept appropriately?**

Validity

EXAMPLE

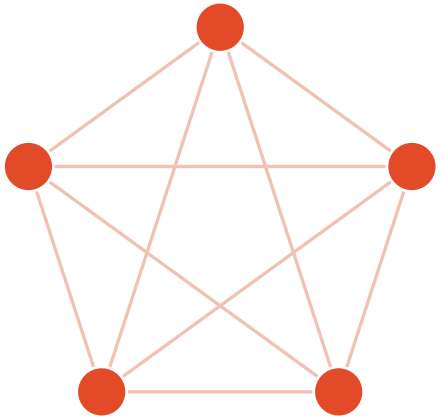
Facebook uses the **clustering coefficient** to recommend friends: if your friends Sanne and Joep aren't connected, Facebook suggests they add each other.



Low clustering

friends aren't friends with each other

*Your measurement of social closure is now measuring **both** social closure and the algorithm's effect. It is algorithmically infused.*



High clustering

friends know each other too

Measurement error

Measurement error: Are your measurements correct? Does the data really capture what happened in reality?

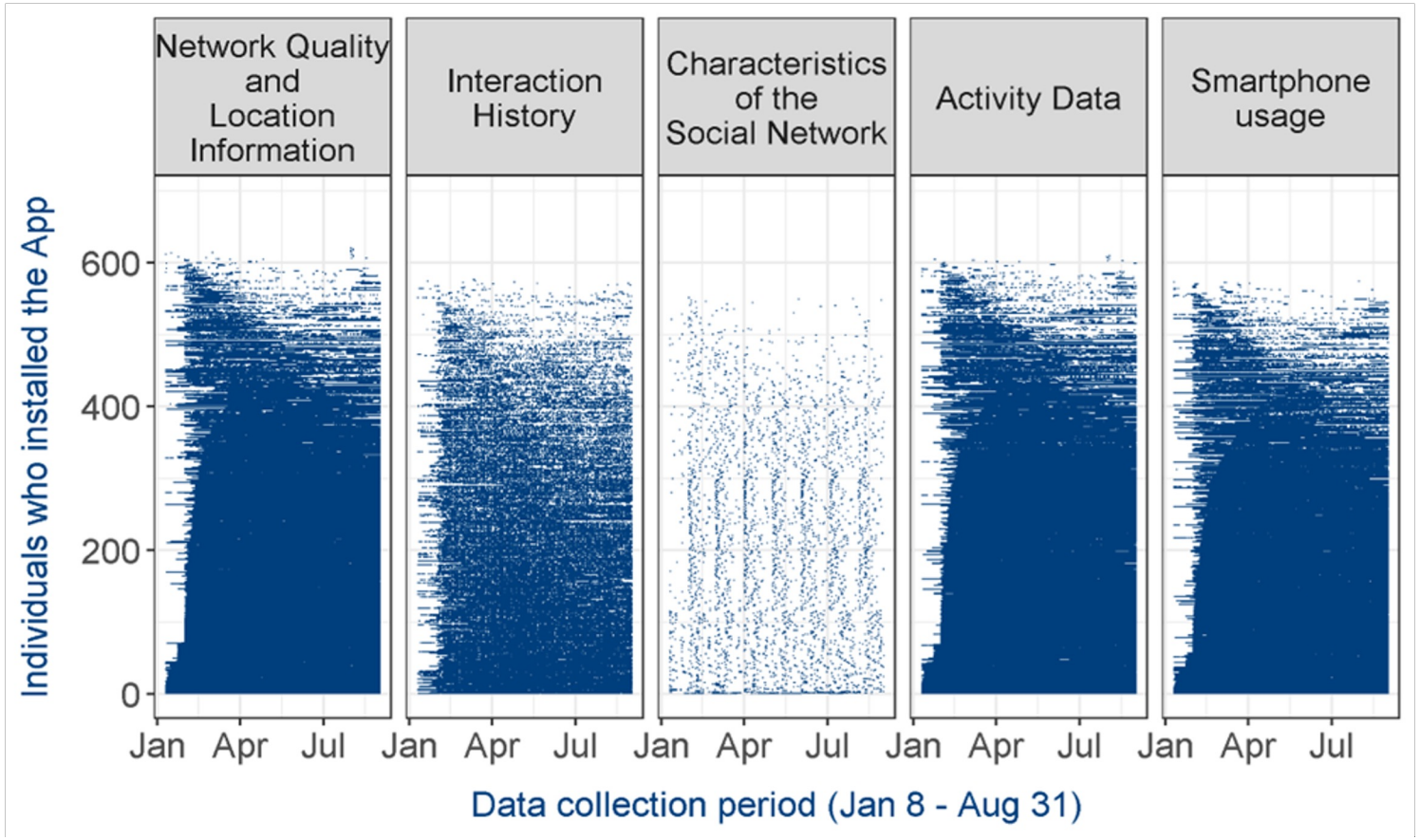
EXAMPLE

- **We are again measuring mood through facial expressions (and now assume it's a good indicator).**
- **We use the Screenomics app.**
- **Is this a correct representation of all your facial expressions, on all your photos?**

Not all devices are tracked, and app/platform use differs by device.
The expressions captured may not represent the ones of the participant

Missing data within a session

- Even once someone agrees to be tracked, sensors and apps don't always capture continuously.
- Gaps can come from battery saving, app permissions, connectivity loss, or the tool not being present at the event.
- The result is a measurement error that varies within a single participant's data, not just across participants.



Bähr et al. (2022)

Platform volatility

EXAMPLE

Change over time: Platforms restructure their Data Download Packages without notice.

Change over OS: Android and Apple exports of the same platform can differ in content and structure.

Differences over language: Field names and structure can shift depending on the user's language setting.

A parser that works today can silently break after the next platform update . The same risk applies to plug-ins and scrapers.

Processing error

Processing error: This error comes in when the raw collected data is cleaned, transformed or converted into the actual measurement or variable that is used for analysis.

In the DTD context, this can happen by multiple processes, e.g:

- When combining multiple traces to obtain measurements on person level
- When classifying the obtained traces into meaningful measurements

Example by T.C. Carrière

1 Validity

2 Measurement error

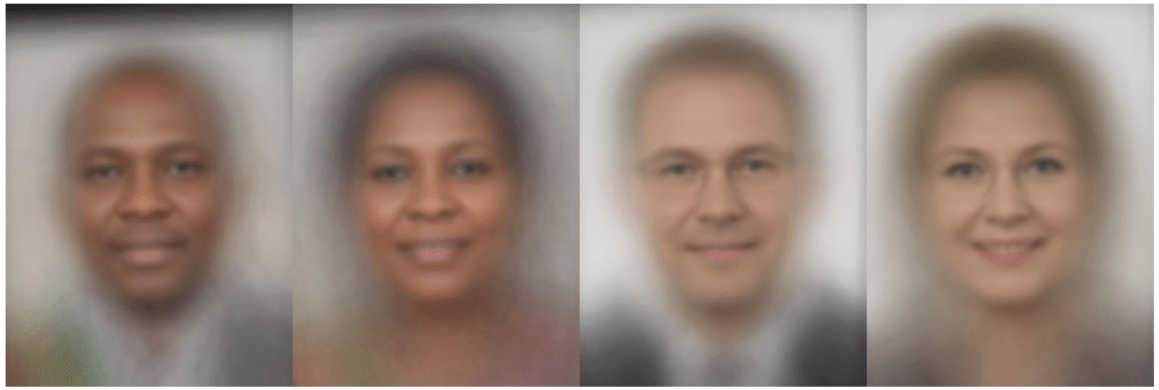
3 **Processing error**

EXAMPLE

Facial recognition software has been shown to work almost perfectly… if you're a white male. Accuracy drops sharply for women and people with darker skin tones.

A processing tool that looks accurate on average can still be badly biased for specific subgroups!

Gender Classifier	Darker Male	Darker Female	Lighter Male	Lighter Female	Largest Gap
Microsoft	94.0%	79.2%	100%	98.3%	20.8%
FACE++	99.3%	65.5%	99.2%	94.0%	33.8%
IBM	88.0%	65.3%	99.7%	92.9%	34.4%



Buolamwini & Gebru (2018), Gender Shades — gendershades.org

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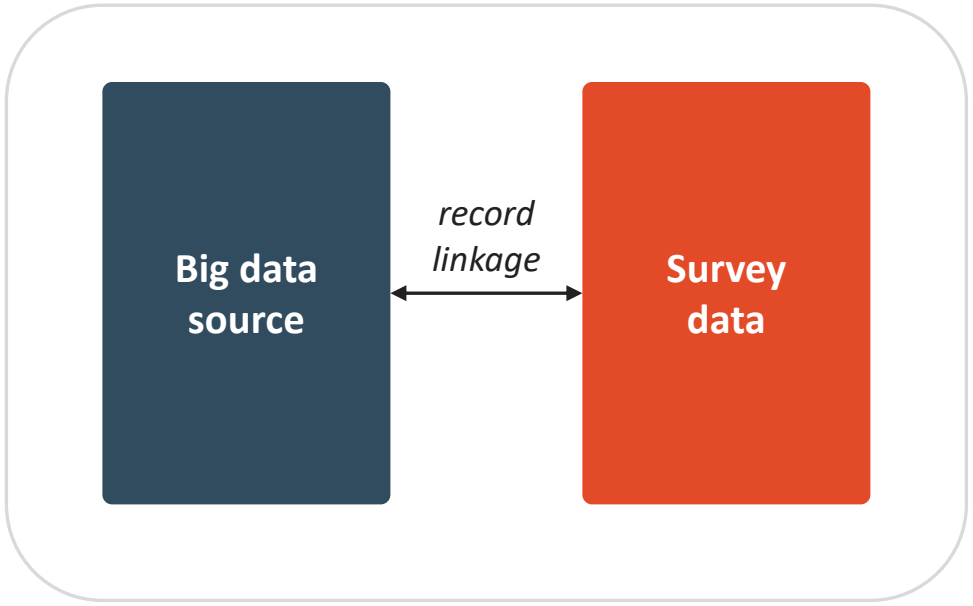
MINUTE COFFEE BREAK

03

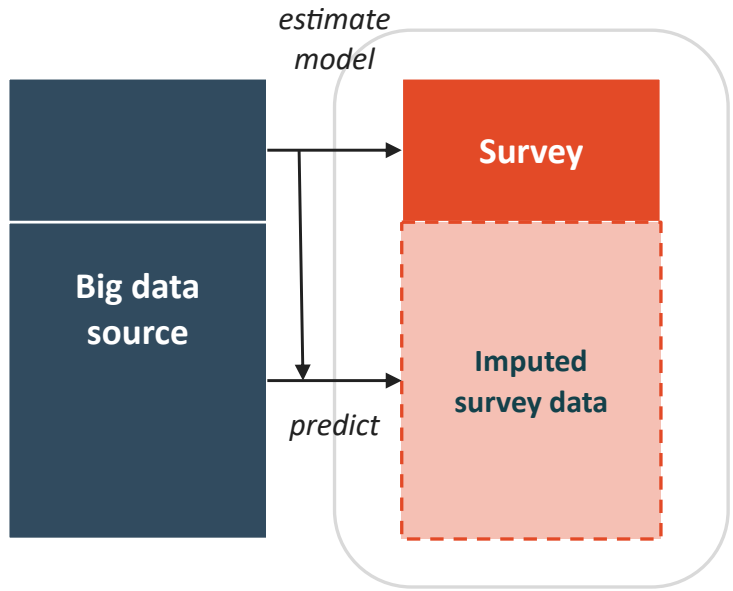
Combining surveys and DTD

Combining surveys and digital trace data

Enriched asking



Amplified asking



Adapted from Salganik (2017), Bit by Bit Fig 3.12

A

Enriched asking

Enriched asking

Enriched asking: The big data source has one measure of interest, and survey data adds other variables of interest for your analysis, for the same people.

- You already have digital traces for a group of people (e.g. Facebook activity, TikTok engagement).
- You add a short survey to that same group to collect variables that are not part of the DTD (attitudes, feelings, context).
- The two sources are linked on person level using record linkage.

Salganik (2017), Bit by Bit §3.6.1

Example: what people say vs. what they do

Research question: *How does what people say about voting differ from their actual voting behavior?*

- The US government records whether each citizen has voted. This is a person-level administrative record.
- This can be supplemented with the attitudes of respondents from a large social survey (the CCES).

Ansolabehere & Hersh (2012)

Building the linked dataset

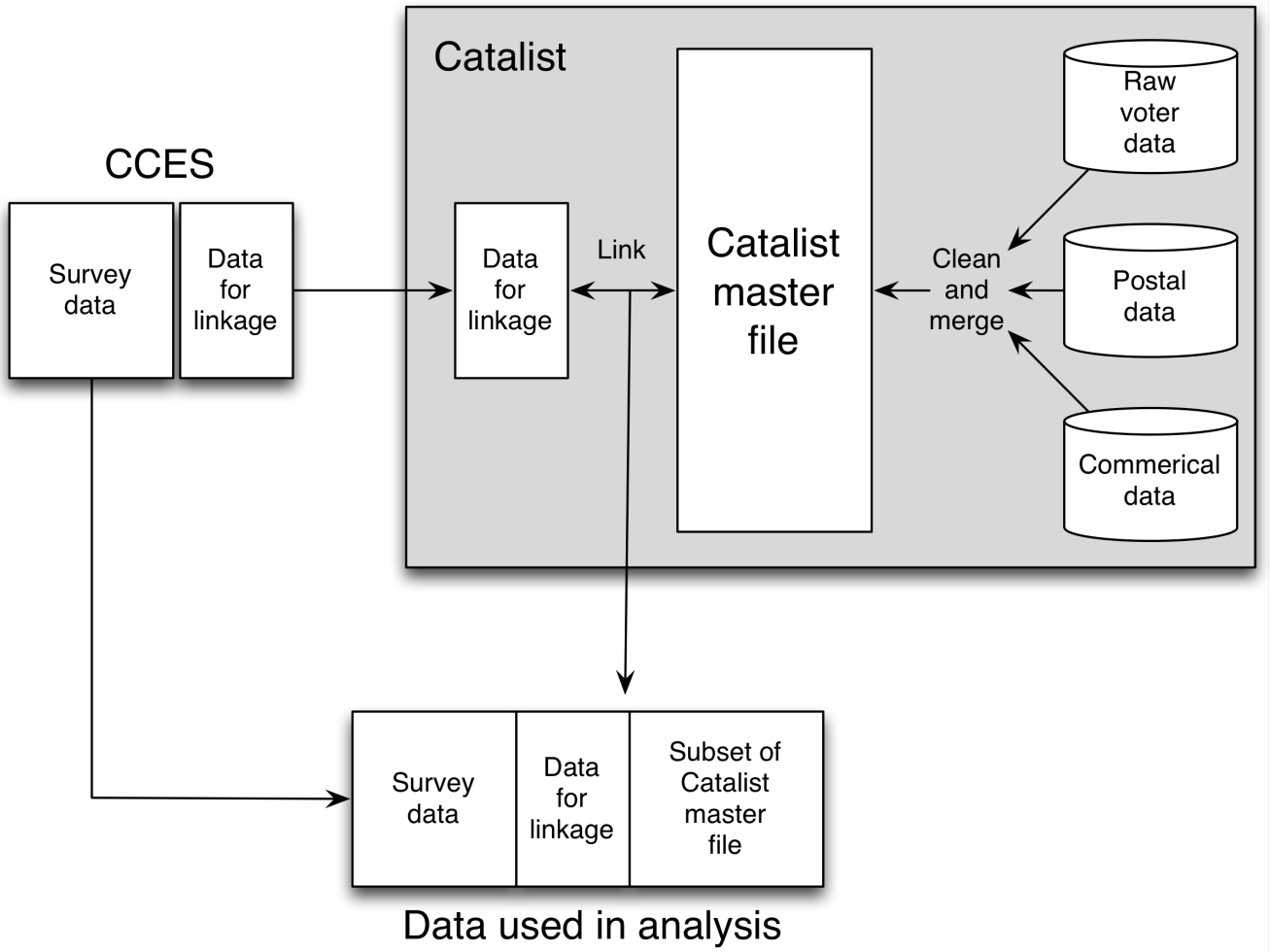


Figure 3.13, Ansolabehere & Hersh (2012), reproduced via Salganik (2017), Bit by Bit

Advantages and challenges

Linking the two sources allows a direct comparison of **self-reported** and **actual (admin)** voting behavior, for the same people.

SOURCES OF ERROR

- The merging done to create the admin Masterfile → combining raw voter data, postal data, and commercial data, each step has its own quality issues.
- No unique identifiers → linkage relies on name, gender, birth year, and address, which can be incomplete, misspelled, or ambiguous (e.g. twins at the same address).

Project results

Public opinion surveys overestimate voter turnout. **If someone reported voting, there's only an 80% chance they actually did.**

Over-reporting isn't random: a particular group consistently misreports, particularly well-educated, high-income, partisan, politically active, church-attending people.

As a result, the real differences between voters and non-voters are smaller than is concluded based on only the survey results.

Ansolabehere & Hersh (2012)

How can enriched asking help?

- There is big value in enriching big data sources, and in enriching surveys → we can do things that aren't possible with just one of the sources.
- Researchers can benefit from the efforts already made by private companies (like Catalist).

Administrative or commercial datasets can't be treated as "ground truth." But surveys also not!

Crucial: record linkage

- Enriched asking assumes perfect linkage between the two data sources.
- If linkage isn't perfect, this can be handled statistically → but that comes with assumptions.
- **Record linkage is essential to making real use of enriched asking.**

EXERCISE

You use Zeeschuimer to scrape 40,000 TikTok posts under #climatecrisis. From this scrape, you identify 500 unique accounts who you invite to complete a survey on eco-anxiety.

What linkage problems do you see?

What assumptions do you need to make?

Suggested answer

Linking a Zeeschuimer TikTok scrape (40,000 posts) to a 500-person eco-anxiety survey via shared handles.

✓ **Potential linkage problems:** Handles can change, can be set to private, or get deleted between scraping and survey time, so exact-string matches can fail. The self-reported handle can have typos. The account might also be shared (e.g. a brand account).

✓ **What assumptions do you need to make:** That the handle someone gave is accurate and still theirs; that the scraped activity reflects their typical behaviour, not a shared or dormant account; and that people who mistype or withhold their handle aren't a systematically different group (e.g. more privacy-conscious).

B

Amplified asking

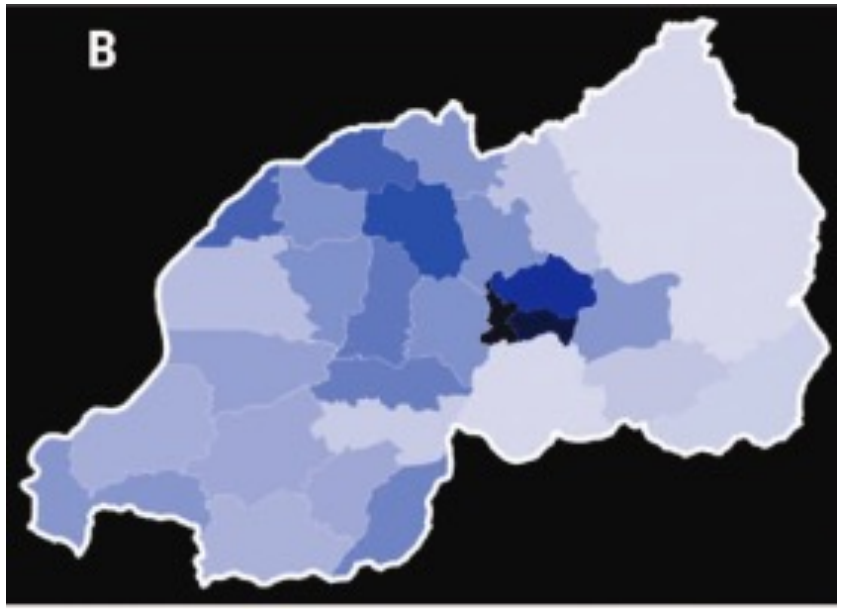
Amplified asking

Amplified asking: Use a predictive model to combine a small amount of survey data with a big data source, to produce estimates at a scale or granularity that would not be possible with either source individually.

Example: study by Blumenstock

Goal: help guide development in poor countries.

Concepts: measure wealth and wellbeing.



Blumenstock et al. (2015)

Traditional approaches

SURVEY

Hard to make estimates about specific geographical regions or demographic groups.

CENSUS

- Expensive.
- Can only ask a few questions.
- Does not happen often.

Combine and get the best of both!

Collaboration with a phone company

Phone company: has all phone record data.

How to supplement with survey data:

1. Select a random sample of phone numbers.
2. Call them.
3. Ask for consent (+ consent to link).
4. Ask survey questions to measure wealth and wellbeing.

Two-step procedure on phone data

Step 1: Feature engineering

Call records converted into a set of characteristics about each person (features / variables). Examples:

- Number of days with activity.
- Number of distinct people a person has been in contact with.
- Amount of money spent on airtime.

Two-step procedure on phone data

Step 2: Supervised learning

- Model to predict the survey response for each person based on their features.
- Used cross-validation to evaluate performance: 10 groups, train on 9, evaluate on the 10th → repeated 10 times.

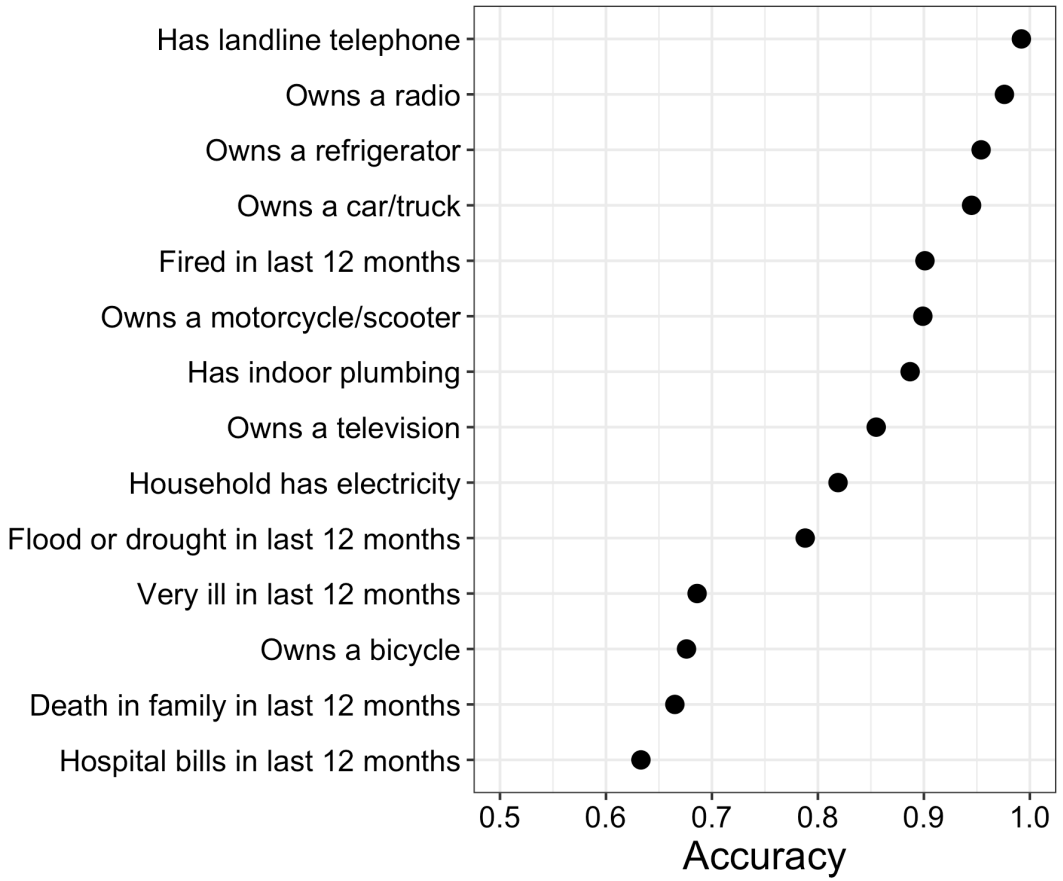
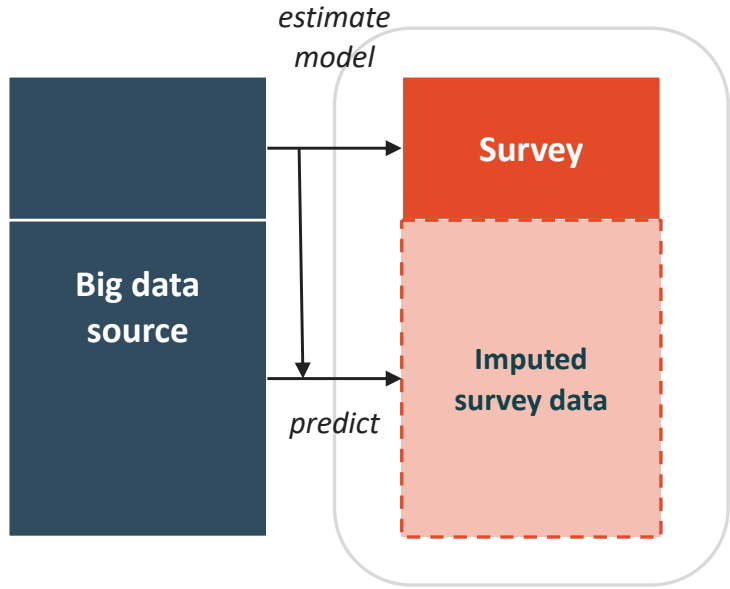


Figure 3.14, adapted from Blumenstock (2014), Table 2

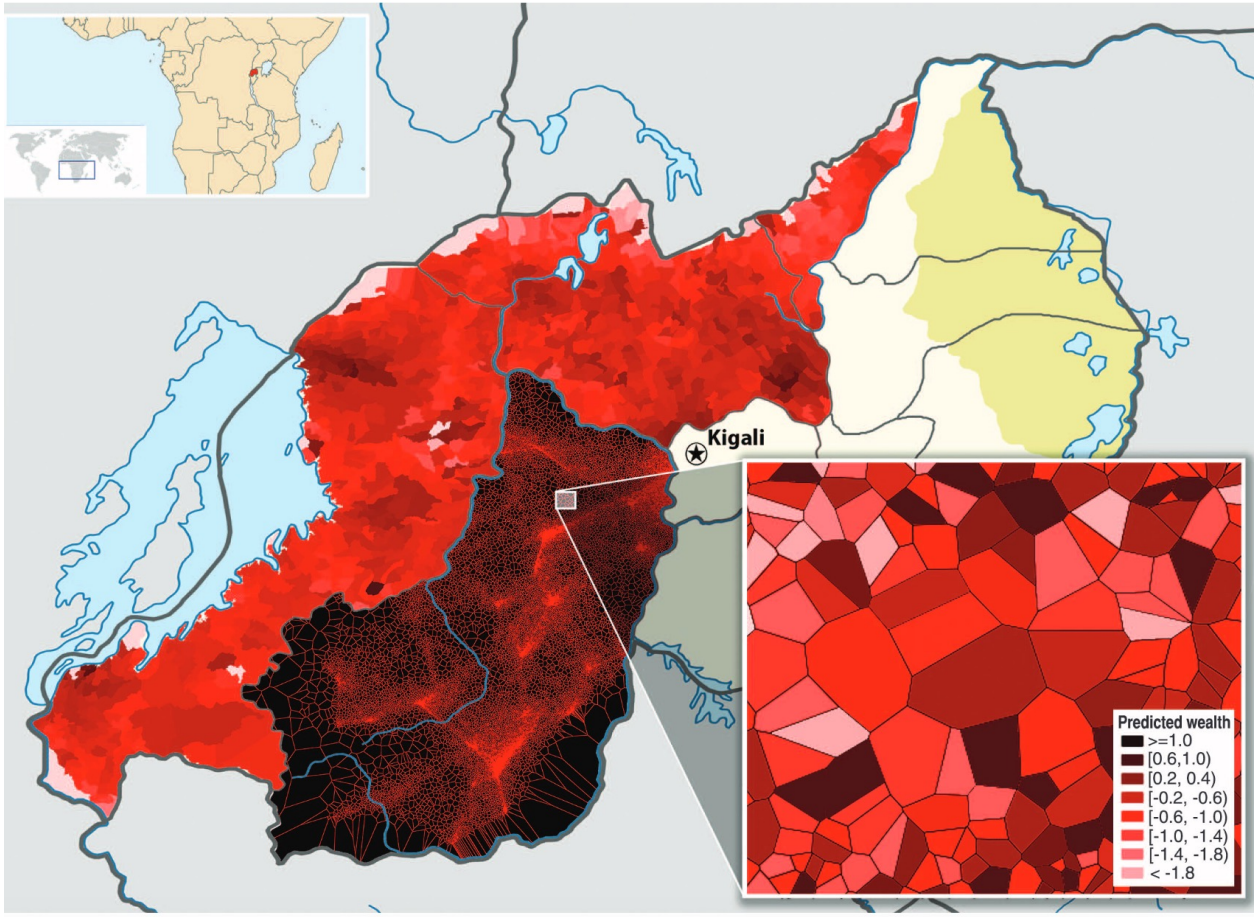
Next: prediction model

- Instead of all separate survey variables, create a composite wealth index (again cross-validated).
- **Predict the wealth index score of all 1.5 million people in the call records.**



Predicting wealth at fine granularity

Geospatial information from the call data gives an estimate of the geographic distribution of wealth at an extremely fine spatial granularity
→ Level of detail: each of Rwanda's neighborhoods.



Blumenstock et al. (2015)

Why be skeptical?

What types of errors are introduced by this procedure?

- Predictions at the individual level are noisy.
- People with mobile phones are systematically different from people without them, especially when it comes to wealth: **coverage error.**
- Errors are introduced during the feature-selection and supervised-learning steps: **algorithmic error for the measurement of wealth.**

Comparison to a high-quality survey

The survey is considered the gold standard, estimates from both approaches were very similar.

10x

faster than the survey

50x

cheaper than the survey

With the budget of a survey that only runs every couple of years, you can do this every month.

Trade-offs: no strong theoretical basis for this kind of approach; we don't know when it will or won't work; and coverage bias especially can be a big issue.

Blumenstock et al. (2015)

General conclusion

When you have:

- One data source with many variables but of few people, and
- Another with few variables of many people …

You can do:

- For the people in both data sources, build a machine learning model that uses the long data source to predict survey answers (wide data source).
- Use that model to infer the survey answers of everyone in the long data source.

Generalize

Can use this approach for many different types of (combinations of) datasets.

E.g. with a user-centric approach, we can:

1. Have many people fill in a survey.
2. Have a small group do the app / donation / etc.
3. Make a model for the survey part.
4. Use that to make predictions for the DTD part.
5. Tailor your survey so it can optimally inform the model.

EXERCISE

You scrape 50,000 Instagram posts with Zeeschuimer, but only 300 of those accounts also filled in your eco-anxiety survey (after agreeing to link their handle).

How would you use amplified asking to estimate wellbeing for all 50,000 accounts? What could make those predictions unreliable?

Suggested answer

Amplifying a 300-person wellbeing survey to estimate wellbeing across 50,000 scraped Instagram accounts.

✓ **How you'd do it :** Engineer features from the scraped posts (posting frequency, caption sentiment, hashtag use, engagement received) for the 300 linked accounts, train a model to predict their survey wellbeing scores from those features, then apply the model to all 50,000 accounts.

✓ **What could make it unreliable:** 300 volunteers who agreed to link their handle are unlikely to represent all 50,000 accounts (coverage/selection bias); individual predictions stay noisy even if the average is accurate; and the choice of features and model introduces its own algorithmic error, with little theoretical basis for knowing when it fails.



Questions?

Laura Boeschoten · Department of Methodology and Statistics